

# 3

SESSION — Oct 15 &amp; 16

## Data Work, and Deterministic Code Over Model Arithmetic

10 · 50 · 10



### GOAL —

Participants stop asking the model to count, compute, or aggregate in its head, and start asking it to write code that does so. Then they learn to read the code.

### WHY —

This is the single best hallucination defense for quantitative work. The model's reasoning is probabilistic; a script is checkable. It also opens code execution to non-programmers, which is where students get the most out of an hour.

### READY-MADE MATERIALS —

Use these when nobody has their own. Everything here is public or was written for this course.

#### The dataset

SIPRI Military Expenditure Database ([sipri.org/databases/milex](https://sipri.org/databases/milex)), the share-of-GDP sheet: one row per country, one column per year. Real, free, and policy-relevant. Fallback if SIPRI is down: the World Bank indicator MS.MIL.XPND.GD.ZS as CSV.

#### The counting question

How many countries spent more than 2% of GDP on the military in the latest year? Ask in prose three times, then Write and run code that counts them, and print the list. Then a chart: the fifteen highest shares as a bar chart.

#### For anyone without their own data

Same file, different question: median share by region, or which countries crossed 2% in the last decade.

## Plan 90 MINUTES

# 10

MIN

### COLD OPEN

Document homework demo, and the failure the pre-flight didn't catch.

50

MIN

CORE

1

**The demonstration 10**

Give the model a spreadsheet of 200 rows and ask "how many rows have value X in column Y?" in plain prose. Then ask it to write and run code that counts. Compare. Do it three times; the prose answer drifts, the code doesn't.

2

**Your own data 20**

Everyone brings a dataset (a CSV, a table pulled from a report, survey results). Task: get one descriptive statistic and one chart via code execution. Non-coders are told to ask the model to explain each line.

3

**Read the code 10**

Breakout pairs. Each person explains their partner's generated script back to them without running it. If you can't explain it, you can't trust it.

4

**Where the trick doesn't apply 10**

Discussion: a lot of RAND work is judgment (framing, synthesis, argument). The code trick fixes arithmetic, not thinking. Which of your tasks this week were arithmetic in disguise? Which were judgment?

10

MIN

DEBRIEF

Log the "count in prose vs. count in code" discrepancy in the Lies Log. Start a "useful snippets" section in the Playbook.

HOMEWORK

Redo one analysis you've already done by hand or in Excel using generated code. Do the numbers match?

60

MIN CUT

FOR 60 MINUTES

Cut step 4 to a two-minute framing statement; fold "read the code" into the debrief.

FACILITATOR  
WATCH-OUTS

- Non-coders will freeze. The fix is the phrase "explain this to me one line at a time." Model it early.
- Someone will get a result that contradicts their Excel. That's the best possible outcome; spend the debrief there.